CS 210: Worksheet 10 *Dr. Mudassir Shabbir, September 29, 2026*

Student ID 1: _____ **Seat Location 1:** row:_____ col:_____

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Mathematical induction and strong induction. For each proof, state the basis, inductive hypothesis, and inductive step, and identify where the hypothesis is used. For strong induction, specify the full range of assumed cases. For the flawed proof, identify the exact error.

1. *Easy*

Use mathematical induction to prove that 3 divides $n^3 + 2n$ for every positive integer n:

$$3 \mid (n^3 + 2n).$$

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2. *Easy*

Use strong induction to prove that every whole-dollar amount of \$4 or more can be formed using only \$2 bills and \$5 bills. Which smaller positive whole-dollar amounts can also be formed?

3. *Easy (guided)*

Let $P(n)$ state that postage of n cents can be formed using only 3-cent and 5-cent stamps. Complete a strong induction proof that $P(n)$ holds for every integer $n \geq 8$.

(a) Verify $P(8)$, $P(9)$, and $P(10)$ to establish the basis.

(b) State the inductive hypothesis, including its range of values.

(c) State what must be proved in the inductive step.

(d) Complete the inductive step for $k \geq 10$.

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4. *Moderate*

Use strong induction to show that every positive integer can be written as a sum of distinct powers of two, chosen from

$$2^0 = 1, \quad 2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8, \quad \ldots.$$

Each power may be used at most once.

Hint: In the inductive step, consider separately whether $k+1$ is even or odd. If it is even, $(k+1)/2$ is an integer.

5. *Moderate*

What is wrong with the following purported proof by strong induction?
Claim. For every nonnegative integer n, $5n = 0$.
Basis. When $n = 0$, $5 \cdot 0 = 0$.
Purported inductive step. Suppose $5j = 0$ for all integers $0 \leq j \leq k$. Write $k + 1 = i + j$, where i and j are nonnegative integers less than $k + 1$. The inductive hypothesis gives

$$5(k + 1) = 5(i + j) = 5i + 5j = 0 + 0 = 0.$$

Identify the first value of k for which this argument fails and explain why.

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6. *Hard (challenge)*

Begin with a pile of n stones, where n is a positive integer. Repeatedly split one pile into two nonempty smaller piles until all n piles contain one stone each. Whenever a pile is split into piles of r and s stones, record the product rs.
Use strong induction to prove that, no matter how you choose the splits, the sum of all recorded products is

$$\frac{n(n-1)}{2}.$$

For $n = 1$, no split is made and the sum is 0.